CTE412 Integrated Systems Technology
Practical 15 – Assessed Lab Report

Item	Detail
Student	Tashi Seldon
Student number (<sid>)	02230153
Programme	BE in Information Technology, Year 3
Cluster / namespace	cte412-02230153 / nb-02230153
Repository	CTE412_Practical15_02230153 (URL submitted on the VLE)
Date of lab session	13 October 2026

1. Objective

The Student Services Office releases new versions of the Campus Notice Board API, and every release so far has caused an unmeasured outage. The purpose of this practical was to replace belief with measurement. I deployed the API to a local three-node Kubernetes cluster, built a continuously running availability monitor that requests the Service twice a second, and then released new versions using three strategies: Recreate, RollingUpdate and blue/green. For each release I counted the failed requests and measured how long old and new versions were served side by side.

I also tested what Kubernetes does with a release that can never become healthy (a broken 1.2.0 image), how a rollback behaves, and whether a feature flag held in a ConfigMap can change behaviour without building a new image. All cluster changes were made declaratively from manifests in the repository; the only imperative change was the rollback in Task 6.3.

2. Environment

Component	Value (evidence)
Operating system	Windows host running Docker Desktop on the WSL2 backend (kernel 6.18.40.1-microsoft-standard-WSL2; t1-nodes-wide.txt). Commands run in PowerShell; shell scripts run with Git for Windows bash.
Docker	Docker Desktop, Linux engine, WSL2 backend; version: 29.8.2 (build 7fc2dff)
kind	kind.exe from C:\kind (version 0.33.0); node image kindest/node:v1.37.0, containerd 2.3.4 (t1-nodes-wide.txt)
kubectl	Client v1.36.1 (within one minor version of the cluster); Kustomize v5.8.1 (t9-versions.txt)
Cluster Kubernetes	Server v1.37.0; one control-plane and two worker nodes, all Ready (t1-nodes-wide.txt)
Application	Python 3.13 (standard-library http.server), built FROM python:3.13-alpine, runs as non-root UID 10001 on port 8080; image size 78.8 MB on disk / 19.5 MB content (t2-local-images.txt)
Monitor image	busybox:1.36 (wget and date), loaded into the cluster with kind load

Every image is Alpine-based or busybox, no Pod pulls a notice-board image from a registry (imagePullPolicy: Never; images loaded with kind load docker-image), and the student number appears in the cluster name, namespace, image tags and every /version response.

3. Results

Request counts and failures come from the monitor logs and the output of scripts/summarise.sh and scripts/window.sh. The monitor requested http://notice-board/version every 0.5 s from inside the cluster and was never restarted. Timestamps have one-second resolution, so windows are accurate to about ±1 s. The "peak Pods" column counts notice-board Pods (not the monitor).

Release	Requests observed	Failed requests	Mixed-version window (s)	Peak notice-board Pods
Recreate: 1.0.0 → 1.1.0 (t5-recreate.log)	113	10	0 (versions never overlapped; 13 s outage 10:32:29–10:32:42Z)	3 (0 Ready during the gap)
RollingUpdate: 1.0.0 → 1.1.0 (t5-rolling.log)	142	0	37 (first 1.1.0 at 10:39:05Z, last 1.0.0 at 10:39:42Z)	4 (3 + 1 surge)
Broken release 1.2.0-broken (t6-monitor.log, whole task)	689	0	0 (broken Pod never became Ready, so 1.2.0-broken was never served)	4 (3 old + 1 broken surge Pod)
Blue/green switch 1: blue → green (13:48:49Z)	467 (one log for all three switches)	0	0 (last 1.0.0 at 13:48:49, first 1.1.0 at 13:48:50)	6 (3 blue + 3 green)
Blue/green switch 2: green → blue (13:49:10Z)	(same log)	0	0 (first 1.0.0 at 13:49:11)	6
Blue/green switch 3: blue → green (13:49:32Z)	(same log)	0	0 (first 1.1.0 at 13:49:32)	6
Flag change FEATURE_PINNED false → true (t8-monitor.log)	2130	0	0 (version stayed 1.1.0 throughout)	7 (3 blue + 3 green + 1 surge)

Other measurements. At rest (Task 4) the monitor recorded 112 requests, 0 failures, all 1.0.0 (t4-summary.txt). The return to 1.0.0 with Recreate (Task 5.2) added a further 8 FAIL lines in the 10:35 minute (checked with the monitor log grouped by minute), giving 18 failures across the two Recreate rollouts. The quota with both colours and the monitor running showed 7 of 15 Pods, 610m of 1 CPU requested and 1550m of 2 CPU limited (t7-quota-both-colours.txt).

Task 6 conditions. After the 60 s deadline Kubernetes reported Progressing = False with reason ProgressDeadlineExceeded, while Available = True with reason MinimumReplicasAvailable (t6-conditions.txt).

4. Analysis

A1. Ranking by failed requests and the mechanism behind each number

Ranking (fewest failures first): RollingUpdate, blue/green, broken release and feature-flag restart all tied at 0 failed requests; Recreate was last with 10 failed requests in the first rollout (t5-recreate-summary.txt) and 8 more in the return rollout.

- **Recreate (10 failures).** The strategy scales the old ReplicaSet to zero before any new Pod exists. In t5-recreate-window.txt the last 1.0.0 response is 10:32:28Z, the first FAIL is 10:32:29Z, the last FAIL is 10:32:42Z and the first 1.1.0 response is 10:32:43Z: a 13 s gap with no ready endpoints. It consists of old-Pod termination, container start, the 5 s warm-up during which /healthz returns 503, and the readiness probe period. A failed wget takes up to its 1 s timeout plus the 0.5 s sleep, so about 10 FAIL lines fit in 13 s. The mixed-version window is 0 s because the versions never overlap.
- **RollingUpdate (0 failures).** maxUnavailable: 0 forbids removing a ready Pod before its replacement is available. The Pod watch (t5-rolling-podwatch.txt) shows the first new Pod appear at 10:38:55 and start running at 10:38:56 as 0/1, while all three old Pods stayed 1/1 Running. The Service always had at least three ready endpoints.
- **Blue/green (0 failures).** Both colours were already Ready, so a switch only changes the Service selector. The EndpointSlice is repointed to Pods that are already serving; no Pod starts or stops. t7-switch-analysis.txt shows the first response from the new colour within one second of each switch and 0 failures in each 20 s window.
- **Broken release (0 failures).** The 1.2.0-broken Pod never passed its readiness probe, so it was never added to the Service endpoints. See A2.
- **Flag restart (0 failures).** The green Deployment was replaced with a RollingUpdate (maxUnavailable 0), so the mechanism is the same as the rolling release. The monitor logged 2130 requests, all 1.1.0 (t8-monitor-summary.txt).

A2. Why the broken release caused no failed requests, and what defaults would have done

The broken image still answers /version, but /healthz always returns 503, so the new Pod stayed 0/1. The Pod list taken 25 s after the apply (t6-pods-during-failure.txt) shows the three old 1.1.0 Pods Running 1/1 plus one new Pod at 0/1, so the old Pods kept serving. The fields responsible, and the predicted result if each had been left at its default:

Field in my manifest	**My value**	**Prediction at the default value**
readinessProbe (httpGet /healthz)	Gates traffic until /healthz is 200	No probe: the Pod is Ready as soon as the container starts and receives traffic. Because /version works on the broken image, my monitor would still record 0 FAIL, but it would show 1.2.0-broken responses and the failure would be invisible to my instrument. With maxUnavailable 0 the rollout would also complete, because the Pod counts as available.
strategy.rollingUpdate.maxUnavailable	0	Default is 25 %. With 3 replicas this rounds down to 0, so the result would be the same here. At 1 or more, an old Pod could be removed while the new one is unready, lowering capacity to 2 Pods during the stall, though still with no failures.
strategy.rollingUpdate.maxSurge	1	Default 25 % rounds up to 1, so the same. With 0 and maxUnavailable 0 the rollout could not start.
minReadySeconds	5	Default 0: a Pod is available the moment it is Ready. No difference for the broken Pod, which is never Ready; it would only matter for a Pod that is Ready briefly and then flaps.
progressDeadlineSeconds	60	Default 600. The traffic outcome would be the same (0 failures), but Kubernetes would take 10 minutes instead of 60 s to report Progressing=False / ProgressDeadlineExceeded, so the failure signal would arrive about 9 minutes later.
livenessProbe (tcpSocket, not /healthz)	TCP check on the named port	If liveness used /healthz, the kubelet would restart the container during the 5 s warm-up and on the broken image, producing restart loops. A TCP check avoids this.

A3. RollingUpdate compared with blue/green

Axis	RollingUpdate (measured)	Blue/green (measured)
Mixed-version window	**37 s** Old and new versions are served together while Pods are replaced one at a time.	0 s in all three switches (t7-switch-analysis.txt). Within one request interval the Service moves from all-blue to all-green. Traffic was never split between versions in the log.
Peak resource use against quota	Peak 4 notice-board Pods (3 + 1 surge). Quota fits comfortably: 4 × 100m = 400m CPU requests and 4 × 250m = 1000m limits, against 1 CPU and 2 CPU.	Peak 6 notice-board Pods plus the monitor = 7 Pods. Measured: requests.cpu 610m of 1, limits.cpu 1550m of 2, requests.memory 592Mi of 1Gi, limits.memory 1184Mi of 2Gi (t7-quota-both-colours.txt). The Task 3 Deployment had to be removed first so that both colours fit.
Time to roll back	Slow: each replacement Pod must start, pass the 5 s warm-up and readiness checks, and stay Ready for minReadySeconds (5 s), so a rollback takes tens of seconds (the Task 6 rollback needed several rollout-status polls).	Fast: switch the Service selector back and apply. Switch 2 (green → blue) at 13:49:10Z had its first 1.0.0 response at 13:49:11Z, about one second.

In short, blue/green buys a near-instant, clean cut-over and rollback at the price of double the Pods. RollingUpdate is cheaper but slower and has a period when two versions answer requests.

A4. A situation where Recreate is the correct choice

Recreate is correct when two versions must never run at the same time. A realistic case is a release that includes a non-backwards-compatible database schema migration, or a single-instance workload holding a ReadWriteOnce volume or an exclusive lock, where version 1.0.0 would break or corrupt data if it were still running while 1.1.0 migrates. The measured 13 s of downtime is acceptable if announced in a maintenance window, because correctness matters more than availability in that case. It is also appropriate when the cluster has no spare quota for a surge Pod or a second colour.

A5. What the feature flag allowed, and what it did not avoid

The flag let me change application behaviour (pinned notices listed first and "pinnedFirst": true) with the same 1.1.0 image: no new build, no new tag and no image load. The ConfigMap value changed to true, and after the flag was active production returned the pinned notices first (t8 evidence; also visible in t9-rebuild.txt).

It did not avoid a rollout. After applying the ConfigMap with FEATURE_PINNED: "true" and waiting 30 s, the three running green Pods still had FEATURE_PINNED=false and production still returned "pinnedFirst": false (t8-flag-observed.txt / E8.2). The variable is injected into the container environment at start-up, so a running process never sees a ConfigMap change. I had to change a Pod-template annotation (cte412/flags-revision: "2") in green.yaml, which forced a RollingUpdate that replaced every green Pod, with 0 failed requests. The flag therefore avoided an image build, but not a Pod restart, and the change still needed a declarative apply.

5. Reflection

An error I met

My first Recreate measurement was invalid. I ran kubectl apply for the "Recreate release" and the output said deployment.apps/notice-board unchanged. The summary then showed 49 requests, 0 failures and only 1.0.0 responses, and the rollout history listed only <none> change-causes (the script also failed with "scripts/window.sh: No such file or directory"). Together these showed that no release had happened. A Select-String of the manifest confirmed it still contained image 1.0.0, version "1.0.0" and no Recreate strategy or change-cause, so my edit had not been written to the file.

The reasoning: kubectl apply compares the manifest with the live object, so "unchanged" means the file I thought I had edited was identical to the cluster state. I deleted the invalid t5-recreate evidence, edited the manifest properly (strategy type Recreate, image 1.1.0, version label 1.1.0 and a change-cause), created scripts/window.sh, and repeated the test. The repeat gave 113 requests and 10 failures, and the rollout history then showed "Release 1.1.0 using Recreate strategy" as revision 3.

One thing that remains unclear

The rollout history shows revisions 1, 4, 6 and 7 rather than a continuous sequence, and the rollback created revision 7 with the change-cause of the earlier 1.1.0 rolling release. I understand that Kubernetes re-uses the number of a ReplicaSet when a rollback re-activates an old template, but I did not verify this in the documentation or the source. Also, because my timestamps have one-second resolution, I cannot tell whether the blue/green switches had a sub-second overlap.

6. References

ConfigMaps. (n.d.). Kubernetes. Retrieved October 13, 2026, from https://kubernetes.io/docs/concepts/configuration/configmap/

Configure Liveness, Readiness and Startup Probes. (n.d.). Kubernetes. Retrieved October 13, 2026, from https://kubernetes.io/docs/tasks/configure-pod-container/configure-liveness-readiness-startup-probes/

Deployments. (n.d.). Kubernetes. Retrieved October 13, 2026, from https://kubernetes.io/docs/concepts/workloads/controllers/deployment/

kind – Quick Start. (2026, July 9). kind. Retrieved October 13, 2026, from https://kind.sigs.k8s.io/docs/user/quick-start/

Resource Quotas. (n.d.). Kubernetes. Retrieved October 13, 2026, from https://kubernetes.io/docs/concepts/policy/resource-quotas/

Service. (2026, July 7). Kubernetes. Retrieved October 13, 2026, from https://kubernetes.io/docs/concepts/services-networking/service/

Appendix – Key Evidence Screenshots

Selected screenshots from the session, each showing the student number 02230153. The full command output is saved as .txt in the evidence/ folder of the repository.

```
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153> kubectl --context kind-cte412-02230153 get nodes -o wide
NAME                           STATUS   ROLES           AGE   VERSION   INTERNAL-IP   EXTERNAL-IP   OS-IMAGE
           KERNEL-VERSION                          CONTAINER-RUNTIME
cte412-02230153-control-plane   Ready    control-plane   54s   v1.37.0   172.20.0.11   <none>        Debian GNU/Linux 1
3 (trixie)   6.18.40.1-microsoft-standard-WSL2 (amd64)   containerd://2.3.4
cte412-02230153-worker          Ready    <none>          39s   v1.37.0   172.20.0.12   <none>        Debian GNU/Linux 1
3 (trixie)   6.18.40.1-microsoft-standard-WSL2 (amd64)   containerd://2.3.4
cte412-02230153-worker2         Ready    <none>          39s   v1.37.0   172.20.0.10   <none>        Debian GNU/Linux 1
3 (trixie)   6.18.40.1-microsoft-standard-WSL2 (amd64)   containerd://2.3.4
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153> kubectl --context kind-cte412-02230153 cluster-info
Kubernetes control plane is running at https://127.0.0.1:54791
CoreDNS is running at https://127.0.0.1:54791/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153>
```

Figure 1 (E1.1): three Ready nodes in wide format, cluster cte412-02230153.

```
Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        10/13/2026   2:34 PM                evidence


PS C:\Users\itech\Desktop\CTE412_Practical15_02230153> Get-ChildItem .\evidence


    Directory: C:\Users\itech\Desktop\CTE412_Practical15_02230153\evidence


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        10/13/2026   2:36 PM           2016 t1-default-namespace.txt
-a----        10/13/2026   2:36 PM           1634 t1-namespace-quota.txt
-a----        10/13/2026   2:24 PM            790 t1-nodes-wide.txt


PS C:\Users\itech\Desktop\CTE412_Practical15_02230153>
```

Figure 2 (E1.2/E1.3): Task 1 evidence files saved in evidence/.

```
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153> C:\kind\kind.exe load docker-image notice-board:02230153-1.0.0 n
otice-board:02230153-1.1.0 notice-board:02230153-1.2.0-broken --name cte412-02230153
>>
>> docker exec cte412-02230153-worker crictl images | Select-String notice-board | Out-File -Encoding ascii evidence\t2
-node-images.txt
>> Get-Content evidence\t2-node-images.txt
Image: "notice-board:02230153-1.0.0" with ID "sha256:19f411786aeeadf7688c8c309f45fdce0842bbecc681f03abae22840a0efce51"
not yet present on node "cte412-02230153-worker", loading...
Image: "notice-board:02230153-1.0.0" with ID "sha256:19f411786aeeadf7688c8c309f45fdce0842bbecc681f03abae22840a0efce51"
not yet present on node "cte412-02230153-worker2", loading...
Image: "notice-board:02230153-1.0.0" with ID "sha256:19f411786aeeadf7688c8c309f45fdce0842bbecc681f03abae22840a0efce51"
not yet present on node "cte412-02230153-control-plane", loading...
Image: "notice-board:02230153-1.1.0" with ID "sha256:60603a2e8920ea5904bb5aa7b0d865bdca8a110917517e4f39b3dfc702a69d61"
not yet present on node "cte412-02230153-worker", loading...
Image: "notice-board:02230153-1.1.0" with ID "sha256:60603a2e8920ea5904bb5aa7b0d865bdca8a110917517e4f39b3dfc702a69d61"
not yet present on node "cte412-02230153-worker2", loading...
Image: "notice-board:02230153-1.1.0" with ID "sha256:60603a2e8920ea5904bb5aa7b0d865bdca8a110917517e4f39b3dfc702a69d61"
not yet present on node "cte412-02230153-control-plane", loading...
Image: "notice-board:02230153-1.2.0-broken" with ID "sha256:9d9325c55ed9752a4ea899e325e7cbd45e8a8305016b923d3841efa711f
571bc" not yet present on node "cte412-02230153-worker", loading...
Image: "notice-board:02230153-1.2.0-broken" with ID "sha256:9d9325c55ed9752a4ea899e325e7cbd45e8a8305016b923d3841efa711f
571bc" not yet present on node "cte412-02230153-worker2", loading...
Image: "notice-board:02230153-1.2.0-broken" with ID "sha256:9d9325c55ed9752a4ea899e325e7cbd45e8a8305016b923d3841efa711f
571bc" not yet present on node "cte412-02230153-control-plane", loading...
```

Figure 3 (E2.2): three release images loaded into the cluster nodes with kind load (no registry).

```
E2.3 startup test, student 02230153
t=0s  /version:   /healthz:  HTTP 000
t=0.6s  /version:   /healthz:  HTTP 000
t=1.3s  /version:   /healthz:  HTTP 000
t=1.9s  /version: {"version": "1.0.0", "student": "02230153", "pod": "3c5bb64bb6f8"}  /healthz: {"status": "unhealthy"}
 HTTP 503
t=2.6s  /version: {"version": "1.0.0", "student": "02230153", "pod": "3c5bb64bb6f8"}  /healthz: {"status": "unhealthy"}
 HTTP 503
t=3.2s  /version: {"version": "1.0.0", "student": "02230153", "pod": "3c5bb64bb6f8"}  /healthz: {"status": "unhealthy"}
 HTTP 503
t=3.8s  /version: {"version": "1.0.0", "student": "02230153", "pod": "3c5bb64bb6f8"}  /healthz: {"status": "unhealthy"}
 HTTP 503
t=4.4s  /version: {"version": "1.0.0", "student": "02230153", "pod": "3c5bb64bb6f8"}  /healthz: {"status": "unhealthy"}
 HTTP 503
t=4.9s  /version: {"version": "1.0.0", "student": "02230153", "pod": "3c5bb64bb6f8"}  /healthz: {"status": "unhealthy"}
 HTTP 503
t=5.5s  /version: {"version": "1.0.0", "student": "02230153", "pod": "3c5bb64bb6f8"}  /healthz: {"status": "unhealthy"}
 HTTP 503
t=6.1s  /version: {"version": "1.0.0", "student": "02230153", "pod": "3c5bb64bb6f8"}  /healthz: {"status": "unhealthy"}
 HTTP 503
t=6.7s  /version: {"version": "1.0.0", "student": "02230153", "pod": "3c5bb64bb6f8"}  /healthz: {"status": "healthy"} H
TTP 200
t=7.3s  /version: {"version": "1.0.0", "student": "02230153", "pod": "3c5bb64bb6f8"}  /healthz: {"status": "healthy"} H
TTP 200
t=7.9s  /version: {"version": "1.0.0", "student": "02230153", "pod": "3c5bb64bb6f8"}  /healthz: {"status": "healthy"} H
TTP 200
t=8.5s  /version: {"version": "1.0.0", "student": "02230153", "pod": "3c5bb64bb6f8"}  /healthz: {"status": "healthy"} H
TTP 200
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153>
```

Figure 4 (E2.3): /version and /healthz from a local 1.0.0 container; 503 unhealthy for the first ~5 s, then 200 healthy.

```
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153> Start-Sleep 5
>> kubectl get pods -l app=notice-board
>> 1..8 | % { curl.exe -s http://127.0.0.1:30090/version; "" }
NAME                            READY   STATUS    RESTARTS   AGE
notice-board-6db6fbf6fb-9m76c   1/1     Running   0          78s
notice-board-6db6fbf6fb-gcpqv   1/1     Running   0          68s
notice-board-6db6fbf6fb-hpp2j   1/1     Running   0          58s
{"version": "1.0.0", "student": "02230153", "pod": "notice-board-6db6fbf6fb-gcpqv"}
{"version": "1.0.0", "student": "02230153", "pod": "notice-board-6db6fbf6fb-hpp2j"}
{"version": "1.0.0", "student": "02230153", "pod": "notice-board-6db6fbf6fb-gcpqv"}
{"version": "1.0.0", "student": "02230153", "pod": "notice-board-6db6fbf6fb-hpp2j"}
{"version": "1.0.0", "student": "02230153", "pod": "notice-board-6db6fbf6fb-hpp2j"}
{"version": "1.0.0", "student": "02230153", "pod": "notice-board-6db6fbf6fb-gcpqv"}
{"version": "1.0.0", "student": "02230153", "pod": "notice-board-6db6fbf6fb-9m76c"}
{"version": "1.0.0", "student": "02230153", "pod": "notice-board-6db6fbf6fb-hpp2j"}
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153>
```

Figure 5 (E3.3): eight /version responses through port 30090, answered by several different Pods.

```
    Conditions:
      Ready:    true
    Hostname:   <unset>
    TargetRef:  Pod/notice-board-6db6fbf6fb-9m76c
    NodeName:   cte412-02230153-worker2
    Zone:       <unset>
  - Addresses:  10.244.2.4
    Conditions:
      Ready:    true
    Hostname:   <unset>
    TargetRef:  Pod/notice-board-6db6fbf6fb-gcpqv
    NodeName:   cte412-02230153-worker
    Zone:       <unset>
  - Addresses:  10.244.1.4
    Conditions:
      Ready:    true
    Hostname:   <unset>
    TargetRef:  Pod/notice-board-6db6fbf6fb-hpp2j
    NodeName:   cte412-02230153-worker2
    Zone:       <unset>
Events:         <none>
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153> git init 2>$null
>> git add .
>> git commit -m "Tasks 1-3: cluster, images, deploy notice-board 1.0.0"
Initialized empty Git repository in C:/Users/itech/Desktop/CTE412_Practical15_02230153/.git/
warning: in the working copy of 'cluster/kind-cluster.yaml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'manifests/base/namespace-quota.yaml', LF will be replaced by CRLF the next time Git to
uches it
warning: in the working copy of 'manifests/rolling/notice-board.yaml', LF will be replaced by CRLF the next time Git to
```

Figure 6 (E3.2): EndpointSlice of notice-board listing three addresses.

```
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153> bash scripts/summarise.sh evidence/t4-at-rest.log | Out-File -En
coding ascii evidence\t4-summary.txt
>> Get-Content evidence\t4-summary.txt
<3>WSL (152081 - Relay) ERROR: CreateProcessCommon:817: execvpe(/bin/bash) failed: No such file or directory
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153> & "C:\Program Files\Git\bin\bash.exe" scripts/summarise.sh evide
nce/t4-at-rest.log | Out-File -Encoding ascii evidence\t4-summary.txt
>> Get-Content evidence\t4-summary.txt
total requests:  112
failed requests: 0
first timestamp: 2026-10-13T10:15:02Z
last timestamp:  2026-10-13T10:16:02Z
responses per version:
  1.0.0: 112
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153> git add -A
>> git commit -m "Task 4: availability monitor, summariser, at-rest evidence"
warning: in the working copy of 'scripts/summarise.sh', LF will be replaced by CRLF the next time Git touches it
[master 45fdac2] Task 4: availability monitor, summariser, at-rest evidence
 5 files changed, 210 insertions(+)
 create mode 100644 evidence/t4-at-rest.log
 create mode 100644 evidence/t4-first10.txt
 create mode 100644 evidence/t4-summary.txt
 create mode 100644 manifests/base/monitor.yaml
 create mode 100644 scripts/summarise.sh
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153>
```

Figure 7 (E4.2): at-rest summary – 112 requests, 0 failed.

```
Waiting for deployment "notice-board" rollout to finish: 0 of 3 updated replicas are available...
Waiting for deployment "notice-board" rollout to finish: 1 of 3 updated replicas are available...
Waiting for deployment "notice-board" rollout to finish: 2 of 3 updated replicas are available...
deployment "notice-board" successfully rolled out
total requests:  113
failed requests: 10
first timestamp: 2026-10-13T10:31:43Z
last timestamp:  2026-10-13T10:32:56Z
responses per version:
  1.1.0: 24
  1.0.0: 79
failed requests:       10
first FAIL:            2026-10-13T10:32:29Z
last FAIL:             2026-10-13T10:32:42Z
failure window:        13 s
first 1.1.0 response: 2026-10-13T10:32:43Z
last 1.0.0 response:  2026-10-13T10:32:28Z
mixed-version window:  0 s (0 = the two versions never overlapped)
deployment.apps/notice-board
REVISION  CHANGE-CAUSE
1         <none>
2         <none>
3         Release 1.1.0 using Recreate strategy

NAME                            READY   STATUS    RESTARTS   AGE
notice-board-79db85f4f4-csc5r   1/1     Running   0          29s
notice-board-79db85f4f4-hdvmr   1/1     Running   0          29s
notice-board-79db85f4f4-xnsbv   1/1     Running   0          29s
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153>
```

Figure 8 (E5.1): Recreate release – 113 requests, 10 failed, 13 s failure window, change-cause recorded.

```
10:38:01 notice-board-6db6fbf6fb-wnf4t   1/1     Running   0          2m30s   10.244.2.10   cte412-02230153-worker    <none>           <none>
10:38:55 notice-board-79db85f4f4-lbh26   0/1     Pending   0          0s      <none>        <none>                    <none>           <none>
10:38:55 notice-board-79db85f4f4-lbh26   0/1     Pending   0          0s      <none>        cte412-02230153-worker2   <none>           <none>
10:38:55 notice-board-79db85f4f4-lbh26   0/1     ContainerCreating   0          0s      <none>        cte412-02230153-worker2   <none>           <n
one>
10:38:56 notice-board-79db85f4f4-lbh26   0/1     ContainerCreating   0          1s      <none>        cte412-02230153-worker2   <none>           <n
one>
10:38:56 notice-board-79db85f4f4-lbh26   0/1     Running             0          1s      10.244.1.7    cte412-02230153-worker2   <none>           <n
one>
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153> Get-Content evidence\t5-rolling-podwatch.txt
10:38:01 NAME                            READY   STATUS    RESTARTS   AGE     IP            NODE                      NOMINATED NODE   READINESS GA
TES
10:38:01 notice-board-6db6fbf6fb-9gm62   1/1     Running   0          2m30s   10.244.1.6    cte412-02230153-worker2   <none>           <none>
10:38:01 notice-board-6db6fbf6fb-b2zfm   1/1     Running   0          2m30s   10.244.2.9    cte412-02230153-worker    <none>           <none>
10:38:01 notice-board-6db6fbf6fb-wnf4t   1/1     Running   0          2m30s   10.244.2.10   cte412-02230153-worker    <none>           <none>
10:38:55 notice-board-79db85f4f4-lbh26   0/1     Pending   0          0s      <none>        <none>                    <none>           <none>
10:38:55 notice-board-79db85f4f4-lbh26   0/1     Pending   0          0s      <none>        cte412-02230153-worker2   <none>           <none>
10:38:55 notice-board-79db85f4f4-lbh26   0/1     ContainerCreating   0          0s      <none>        cte412-02230153-worker2   <none>           <n
one>
10:38:56 notice-board-79db85f4f4-lbh26   0/1     ContainerCreating   0          1s      <none>        cte412-02230153-worker2   <none>           <n
one>
10:38:56 notice-board-79db85f4f4-lbh26   0/1     Running             0          1s      10.244.1.7    cte412-02230153-worker2   <none>           <n
one>
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153>
```

Figure 9 (E5.4): Pod watch during the RollingUpdate release (new Pod created while old Pods remain Running).

```
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153> $since6 = (Get-Date).ToUniversalTime().AddSeconds(-2).ToString("yyyy-MM-ddTHH:mm:ssZ")
>> $since6 | Out-File -Encoding ascii evidence\t6-since.txt
>> Start-Sleep 12
>> kubectl apply -f $f
>> Start-Sleep 25
>> kubectl get pods -l app=notice-board -o wide | Out-File -Encoding ascii evidence\t6-pods-during-failure.txt
NAME                            READY   STATUS    RESTARTS   AGE     IP            NODE                      NOMINATED NODE   READINESS GATES
notice-board-66b9c8569b-959vt   0/1     Running   0          26s     10.244.2.12   cte412-02230153-worker    <none>           <none>
notice-board-79db85f4f4-lbh26   1/1     Running   0          7m10s   10.244.1.7    cte412-02230153-worker2   <none>           <none>
notice-board-79db85f4f4-prbbx   1/1     Running   0          6m55s   10.244.2.11   cte412-02230153-worker    <none>           <none>
notice-board-79db85f4f4-r6vcm   1/1     Running   0          6m39s   10.244.1.8    cte412-02230153-worker2   <none>           <none>
kubectl : error: deployment "notice-board" exceeded its progress deadline
At line:9 char:1
+ kubectl rollout status deployment/notice-board --timeout=10s 2>&1 | O ...
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : NotSpecified: (error: deployme...ogress deadline:String) [], RemoteException
    + FullyQualifiedErrorId : NativeCommandError



type    : Available
status  : True
reason  : MinimumReplicasAvailable
message : Deployment has minimum availability.

type    : Progressing
status  : False
reason  : ProgressDeadlineExceeded
message : ReplicaSet "notice-board-66b9c8569b" has timed out progressing.



PS C:\Users\itech\Desktop\CTE412_Practical15_02230153>
```

Figure 10 (E6.1/E6.2): broken release – old Pods still serving; Progressing=False, ProgressDeadlineExceeded.

```
>> kubectl get pods -l app=notice-board -o wide
>> kubectl rollout history deployment/notice-board
Warning: resource deployments/notice-board was previously managed with 'kubectl apply'. Rolling back will not update the kubectl.kubernetes.io/last
-applied-configuration annotation, which may cause unexpected behavior on future 'kubectl apply' operations. Consider using 'kubectl apply' with yo
ur previous configuration file instead.
deployment.apps/notice-board rolled back
Waiting for deployment "notice-board" rollout to finish: 1 old replicas are pending termination...
deployment "notice-board" successfully rolled out
total requests:  328
failed requests: 0
first timestamp: 2026-10-13T10:45:25Z
last timestamp:  2026-10-13T10:48:16Z
responses per version:
  1.1.0: 328
NAME                            READY   STATUS        RESTARTS   AGE     IP            NODE                       NOMINATED NODE   READINESS GATES
notice-board-66b9c8569b-959vt   0/1     Terminating   0          2m38s   10.244.2.12   cte412-02230153-worker     <none>           <none>
notice-board-79db85f4f4-lbh26   1/1     Running       0          9m22s   10.244.1.7    cte412-02230153-worker2    <none>           <none>
notice-board-79db85f4f4-prbbx   1/1     Running       0          9m7s    10.244.2.11   cte412-02230153-worker     <none>           <none>
notice-board-79db85f4f4-r6vcm   1/1     Running       0          8m51s   10.244.1.8    cte412-02230153-worker2    <none>           <none>
deployment.apps/notice-board
REVISION  CHANGE-CAUSE
1         <none>
4         Return to 1.0.0 using Recreate strategy
6         Release 1.2.0-broken (expected to fail its readiness probe)
7         Release 1.1.0 using RollingUpdate (maxUnavailable 0, maxSurge 1, minReadySeconds 5)

PS C:\Users\itech\Desktop\CTE412_Practical15_02230153>
```

Figure 11 (E6.3): rollback and monitor summary with 0 failed requests.

```
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153> $f = ".\manifests\rolling\notice-board.yaml"
>> "E6.4 manifest vs live object, after rollback and re-apply" | Out-File -Encoding ascii evidence\t6-manifest-agreement.t
xt
>> "--- dry-run apply (every object must say unchanged) ---" | Out-File -Encoding ascii evidence\t6-manifest-agreement.txt
 -Append
>> kubectl apply --dry-run=server -f $f 2>&1 | Out-File -Encoding ascii evidence\t6-manifest-agreement.txt -Append
>> "--- live Deployment fields ---" | Out-File -Encoding ascii evidence\t6-manifest-agreement.txt -Append
>> $live = kubectl get deployment notice-board -o json | ConvertFrom-Json
>> "image=$($live.spec.template.spec.containers[0].image) version=$($live.spec.template.metadata.labels.version) progressD
eadlineSeconds=$($live.spec.progressDeadlineSeconds) minReadySeconds=$($live.spec.minReadySeconds)" | Out-File -Encoding a
scii evidence\t6-manifest-agreement.txt -Append
>> "--- same fields in the manifest ---" | Out-File -Encoding ascii evidence\t6-manifest-agreement.txt -Append
>> Select-String -Path $f -Pattern "progressDeadlineSeconds|minReadySeconds|image:|version: `"" | ForEach-Object { $_.Line
.Trim() } | Out-File -Encoding ascii evidence\t6-manifest-agreement.txt -Append
>> Get-Content evidence\t6-manifest-agreement.txt
E6.4 manifest vs live object, after rollback and re-apply
--- dry-run apply (every object must say unchanged) ---
deployment.apps/notice-board unchanged (server dry run)
service/notice-board unchanged (server dry run)
service/notice-board-ext unchanged (server dry run)
--- live Deployment fields ---
image=notice-board:02230153-1.1.0 version=1.1.0 progressDeadlineSeconds=60 minReadySeconds=5
--- same fields in the manifest ---
kubernetes.io/change-cause: "Release 1.1.0 using RollingUpdate (maxUnavailable 0, maxSurge 1, minReadySeconds 5)"
progressDeadlineSeconds: 60
minReadySeconds: 5
version: "1.1.0"
image: notice-board:02230153-1.1.0
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153>
```

Figure 12 (E6.4): manifest and live Deployment agree after the rollback (dry-run apply: unchanged).

```
>>
>> # E7.4: quota Used column with both colours AND the monitor running
>> kubectl describe quota | Out-File -Encoding ascii evidence\t7-quota-both-colours.txt
>> Get-Content evidence\t7-quota-both-colours.txt
--- preview (green) ---
{"version": "1.1.0", "student": "02230153", "pod": "notice-board-green-57687f5467-2gphs"}
{"version": "1.1.0", "student": "02230153", "pod": "notice-board-green-57687f5467-2gphs"}
{"version": "1.1.0", "student": "02230153", "pod": "notice-board-green-57687f5467-2gphs"}
{"version": "1.1.0", "student": "02230153", "pod": "notice-board-green-57687f5467-2rqtp"}
--- production (blue) ---
{"version": "1.0.0", "student": "02230153", "pod": "notice-board-blue-66f69c6cc4-h22g8"}
{"version": "1.0.0", "student": "02230153", "pod": "notice-board-blue-66f69c6cc4-d6xpc"}
{"version": "1.0.0", "student": "02230153", "pod": "notice-board-blue-66f69c6cc4-svm7d"}
{"version": "1.0.0", "student": "02230153", "pod": "notice-board-blue-66f69c6cc4-svm7d"}
Name:            nb-quota
Namespace:       nb-02230153
Resource         Used    Hard
--------         ----    ----
count/services   3       4
limits.cpu       1550m   2
limits.memory    1184Mi  2Gi
pods             7       15
requests.cpu     610m    1
requests.memory  592Mi   1Gi
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153>
```

Figure 13 (E7.2/E7.4): preview returns 1.1.0 while production returns 1.0.0; quota Used column with both colours running.

```
deployment "notice-board-green" successfully rolled out
NAME                               READY   UP-TO-DATE   AVAILABLE   AGE   CONTAINERS     IMAGES
                SELECTOR                      LABELS
deployment.apps/notice-board-blue    3/3     3            3           20s   notice-board   notice-board:
02230153-1.0.0   app=notice-board,slot=blue    <none>
deployment.apps/notice-board-green   3/3     3            3           19s   notice-board   notice-board:
02230153-1.1.0   app=notice-board,slot=green   <none>

NAME                                    READY   STATUS    RESTARTS   AGE   IP           NODE
           NOMINATED NODE   READINESS GATES   LABELS
pod/notice-board-blue-66f69c6cc4-d6xpc    1/1     Running   0          19s   10.244.2.2   cte412-0223015
3-worker    <none>           <none>            app=notice-board,pod-template-hash=66f69c6cc4,slot=blue,v
ersion=1.0.0
pod/notice-board-blue-66f69c6cc4-h22g8    1/1     Running   0          19s   10.244.3.2   cte412-0223015
3-worker2   <none>           <none>            app=notice-board,pod-template-hash=66f69c6cc4,slot=blue,v
ersion=1.0.0
pod/notice-board-blue-66f69c6cc4-svm7d    1/1     Running   0          19s   10.244.3.3   cte412-0223015
3-worker2   <none>           <none>            app=notice-board,pod-template-hash=66f69c6cc4,slot=blue,v
ersion=1.0.0
pod/notice-board-green-57687f5467-2gphs   1/1     Running   0          19s   10.244.3.4   cte412-0223015
3-worker2   <none>           <none>            app=notice-board,pod-template-hash=57687f5467,slot=green,
version=1.1.0
pod/notice-board-green-57687f5467-2rqtp   1/1     Running   0          19s   10.244.2.4   cte412-0223015
3-worker    <none>           <none>            app=notice-board,pod-template-hash=57687f5467,slot=green,
version=1.1.0
```

Figure 14 (E7.1): blue and green Pods with slot labels, and Services selecting slot=blue (production) and slot=green (preview).

```
>>   } else { $mixed = 0; $loS = 'none after switch' }
>>   "$t $label  first-$new=$fnS  last-$old=$loS  mixed-window=${mixed}s  fails=$fails"
>> }
>> $res | Tee-Object evidence\t7-switch-analysis.txt
>>
>> # Marked log, fixed array handling (ISO timestamps sort correctly as strings)
>> $out = New-Object System.Collections.Generic.List[string]
>> $i = 0
>> foreach ($line in (Get-Content evidence\t7-monitor.log)) {
>>   while ($i -lt $marks.Count -and $line -match '^\d{4}-' -and ($line -split ' ')[0] -ge ($marks[$i] -split ' ')[0]) {
>>     $out.Add("### SWITCH MARK: $($marks[$i])"); $i++
>>   }
>>   $out.Add($line)
>> }
>> $out | Out-File -Encoding ascii evidence\t7-monitor-marked.log
>> Select-String -Path evidence\t7-monitor-marked.log -Pattern "SWITCH MARK"
>>
>> git add -A
>> git commit -m "Task 7: fix marked log and null-safe mixed-window analysis"
2026-10-13T13:48:49Z switch-to-green  first-1.1.0=13:48:50  last-1.0.0=13:48:49  mixed-window=0s  fails=0
2026-10-13T13:49:10Z switch-to-blue  first-1.0.0=13:49:11  last-1.1.0=13:49:11  mixed-window=0s  fails=0
2026-10-13T13:49:32Z switch-to-green  first-1.1.0=13:49:32  last-1.0.0=none after switch  mixed-window=0s  fails=0

evidence\t7-monitor-marked.log:348:### SWITCH MARK: 2026-10-13T13:48:49Z switch-to-green
evidence\t7-monitor-marked.log:388:### SWITCH MARK: 2026-10-13T13:49:10Z switch-to-blue
evidence\t7-monitor-marked.log:430:### SWITCH MARK: 2026-10-13T13:49:32Z switch-to-green
[master 9d1c628] Task 7: fix marked log and null-safe mixed-window analysis
 2 files changed, 1 insertion(+), 2 deletions(-)
```

Figure 15 (E7.3): per-switch analysis and switch marks in the monitor log (0 failures, 0 s mixed window).

```
>>   $out += "$p = $(kubectl exec $p -- printenv FEATURE_PINNED)"
>> }
>> $out += ""
>> $out += "Production /notices after the flag change, before any restart:"
>> $out += (curl.exe -s http://127.0.0.1:30090/notices)
>> $out | Tee-Object evidence\t8-flag-observed.txt
pod/notice-board-green-6cc8d879fb-k2mcl = false
pod/notice-board-green-6cc8d879fb-mt88m = false
pod/notice-board-green-6cc8d879fb-zqfk4 = false

manifests\bluegreen\flags.yaml:11:  FEATURE_PINNED: "true"   # true = pinned notices listed first (flag flipped in Task 8)
configmap/notice-board-flags unchanged
ConfigMap value:
true

Env inside each running green Pod:
pod/notice-board-green-6cc8d879fb-k2mcl = false
pod/notice-board-green-6cc8d879fb-mt88m = false
pod/notice-board-green-6cc8d879fb-zqfk4 = false

Production /notices after the flag change, before any restart:
{"notices": [{"id": 1, "title": "Library Hours", "body": "The library opens at 8 AM.", "category": "Facilities", "pinned":
true}, {"id": 2, "title": "Campus Meeting", "body": "The campus meeting is on Friday.", "category": "Events", "pinned": fal
se}, {"id": 3, "title": "Exam Schedule", "body": "Check the examination timetable.", "category": "Examinations", "pinned":
true}], "pinnedFirst": false}


PS C:\Users\itech\Desktop\CTE412_Practical15_02230153>
```

Figure 16 (E8.2): ConfigMap is true but running green Pods still report FEATURE_PINNED=false; production pinnedFirst is false.

```
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153> cd C:\Users\itech\Desktop\CTE412_Practical15_02230153
>>
>> git diff manifests/bluegreen/green.yaml | Select-String -Pattern 'annotations|flags-revision'
>>
>> kubectl logs monitor --since-time=2026-10-13T13:56:45Z | Out-File -Encoding ascii evidence\t8-monitor.log
>> & "C:\Program Files\Git\bin\bash.exe" scripts/summarise.sh evidence/t8-monitor.log | Tee-Object evidence\t8-monitor-summ
ary.txt

+      annotations:
+        cte412/flags-revision: "2"
total requests:  2130
failed requests: 0
first timestamp: 2026-10-13T13:56:45Z
last timestamp:  2026-10-13T14:16:46Z
responses per version:
  1.1.0: 2130


PS C:\Users\itech\Desktop\CTE412_Practical15_02230153>
```

Figure 17 (E8.3): Task 8 monitor summary – 2130 requests, 0 failed.

```
deployment.apps/notice-board-blue    3/3     3            3           19s
deployment.apps/notice-board-green   3/3     3            3           19s

NAME                                          DESIRED   CURRENT   READY   AGE
replicaset.apps/notice-board-blue-66f69c6cc4   3         3         3       19s
replicaset.apps/notice-board-green-564f87986d  3         3         3       19s

NAME                         DATA   AGE
configmap/kube-root-ca.crt   1      52s
configmap/notice-board-flags 1      19s

NAME                                                ADDRESSTYPE   PORTS   ENDPOINTS                         AGE
endpointslice.discovery.k8s.io/notice-board-ext-bhplc       IPv4          8080    10.244.1.6,10.244.2.8,10.244.1.7   20s
endpointslice.discovery.k8s.io/notice-board-preview-mlvfd   IPv4          8080    10.244.1.6,10.244.2.8,10.244.1.7   20s
endpointslice.discovery.k8s.io/notice-board-xl99l           IPv4          8080    10.244.1.6,10.244.2.8,10.244.1.7   20s
{"version": "1.1.0", "student": "02230153", "pod": "notice-board-green-564f87986d-dbgtd"}{"notices": [{"id": 1, "title": "L
ibrary Hours", "body": "The library opens at 8 AM.", "category": "Facilities", "pinned": true}, {"id": 3, "title": "Exam Sc
hedule", "body": "Check the examination timetable.", "category": "Examinations", "pinned": true}, {"id": 2, "title": "Campu
s Meeting", "body": "The campus meeting is on Friday.", "category": "Events", "pinned": false}], "pinnedFirst": true}Transc
ript stopped, output file is C:\Users\itech\Desktop\CTE412_Practical15_02230153\evidence\t9-rebuild.txt
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153> git add evidence/t9-rebuild.txt evidence/t9-versions.txt
>> git commit -m "T9: rebuild proof on fresh cluster"
warning: in the working copy of 'evidence/t9-rebuild.txt', LF will be replaced by CRLF the next time Git touches it
[master 3e2f1e8] T9: rebuild proof on fresh cluster
 2 files changed, 71 insertions(+)
 create mode 100644 evidence/t9-rebuild.txt
 create mode 100644 evidence/t9-versions.txt
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153>
```

Figure 18 (E9.1): rebuild on a fresh cluster from the repository; pinnedFirst is true.

```
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153> Get-Content evidence\t9-versions.txt
Client Version: v1.36.1
Kustomize Version: v5.8.1
Server Version: v1.37.0
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153> Start-Transcript -Path evidence\t9-cluster-deleted.txt
>> kubectl config set-context --current --namespace=default
>> kind delete cluster --name cte412-02230153
>> kind get clusters
>> docker ps -a --filter "name=cte412-02230153"
>> kubectl config get-contexts
>> Stop-Transcript
Transcript started, output file is evidence\t9-cluster-deleted.txt
Context "kind-cte412-02230153" modified.
Deleting cluster "cte412-02230153" ...
Deleted nodes: ["cte412-02230153-control-plane" "cte412-02230153-worker" "cte412-02230153-worker2"]
4it
dso202
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
CURRENT   NAME             CLUSTER          AUTHINFO         NAMESPACE
          docker-desktop   docker-desktop   docker-desktop
          kind-4it         kind-4it         kind-4it
          kind-dso202      kind-dso202      kind-dso202      dso202-practical
          minikube         minikube         minikube         default
Transcript stopped, output file is C:\Users\itech\Desktop\CTE412_Practical15_02230153\evidence\t9-cluster-deleted.txt
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153>
```

Figure 19 (E9.2): context reset to default and cluster cte412-02230153 deleted.

```
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153> git remote add origin https://github.com/TashiSeldon/CTE412_Practica
l15_02230153.git
>> git branch --show-current
>> git status
>> git log --oneline
master
On branch master
nothing to commit, working tree clean
2d3dadf (HEAD -> master) T9: rebuild proof, cluster deleted, final evidence
3e2f1e8 T9: rebuild proof on fresh cluster
1a814aa T9: assembled monitor log from per-task logs
be7c334 README: rebuild sequence, versions, clean-up
99dfd77 Task 8: feature flag ConfigMap, env from ConfigMap, flag rollout evidence
9d1c628 Task 7: fix marked log and null-safe mixed-window analysis
7c90576 Task 7: summary, mixed-version windows, marked monitor log
45cf513 Task 7: blue/green switch evidence, monitor summary and per-switch analysis
6239977 Task 7: blue/green manifests, pre-switch evidence
6d6b527 Task 6: manifest reconciled with live Deployment after rollback
d664f7c Task 6: broken release, rollback, zero-failure monitor evidence
530f2f2 Task 5: complete evidence
313cd3d Task 5: RollingUpdate release 1.1.0 with monitor and pod watch evidence
d0de028 Task 5: return to 1.0.0 declaratively
396d8f7 Task 5: Recreate release 1.1.0 with monitor evidence
45fdac2 Task 4: availability monitor, summariser, at-rest evidence
bafbd6c Remove stale evidence files superseded by corrected app
a9ab258 Tasks 1-3: cluster, images, deploy notice-board 1.0.0
PS C:\Users\itech\Desktop\CTE412_Practical15_02230153>   git add -A
>>   git commit -m "Final evidence, report and clean-up"
```

Figure 20 (repository): git history of the repository, 18 commits made during the lab session, one or more per task (git log --oneline).